

Speech Enhancement Using Wiener Filtering and Short-Time Fourier Transform (STFT)

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Project Overview

Speech communication is often affected by white noise especially from the environment, which will reduce speech quality and intelligibility, particularly in noisy environments such as streets, public transport, and crowded indoor spaces [1]. Speech enhancement techniques aim to reduce those noise while preserving the important components of the speech signal [1], [2].

The aim of this project is to develop a speech enhancement system using Short-Time Fourier Transform (STFT) and Wiener filtering. The noisy speech signal will first be transformed into the time-frequency domain using STFT, allowing the spectral characteristics of speech and noise to be analysed over short time frames [2]. Wiener filtering will then be applied in the STFT domain to reduce noise components while preserving the desired speech signal [1], [2]. Finally, the enhanced speech signal will be reconstructed using inverse STFT. The performance of the proposed system will be evaluated by comparing the noisy and enhanced speech signals using Signal-to-Noise Ratio (SNR).

Background and Motivation

Speech enhancement is an important area of digital signal processing because speech signals are often affected by background noise in real-world environments. This noise will reduce speech quality and make speech more difficult to understand, especially in applications such as mobile communication, voice recording, hearing devices, and speech recognition systems [1]. Therefore, reducing effective noise is important for improving the quality and reliability of speech communication.

Several traditional methods have been developed for speech enhancement, which includes spectral subtraction, Wiener filtering and minimum mean-square error (MMSE) estimation [1], [3]. Among these methods, Wiener filtering is most widely used to recover clean speech from noisy signals. It reduces the difference between the estimated speech and the desired clean speech, providing a simple and effective method for noise reduction [2].

Speech signals change over time, so they can be analysed in short time frames using the Short-Time Fourier Transform (STFT). STFT converts the speech signal into a time-frequency representation, allowing noise reduction methods to process different frequency components [2], [3]. SNR estimation is also important in frequency-domain speech enhancement and can help evaluate the performance of the enhancement system [5].

This project will combine STFT and Wiener filtering to improve noisy speech signals. Publicly available datasets such as LibriSpeech provide clean speech samples that can be used to develop and test the proposed system [4]. It provides a practical way to apply important digital signal processing concepts and allows the performance of the system to be tested and compared under different types and levels of background noise.

Proposed Methodology

Tools and Platforms

- MATLAB

Signal Processing Techniques

- Short-Time Fourier Transform (STFT)
- Wiener Filtering
- Inverse Short-Time Fourier Transform (ISTFT)
- Signal-to-Noise Ratio (SNR)

Data Sources

- Mini LibriSpeech / LibriSpeech
- Different types of background noise will be added to the clean speech signals to generate noisy speech samples at different SNR levels.

Expected Outcomes

The expected outcomes of this project include:

- A working MATLAB-based speech enhancement system using STFT and Wiener filtering to reduce background noise from speech signals.
- Enhanced speech signals with reduced background noise compared with the original noisy speech.
- Performance evaluation using SNR, including comparison of SNR before and after speech enhancement.
- Waveform and spectrogram comparisons to visually demonstrate the effect of noise reduction.
- Experimental results under different noise levels to evaluate the effectiveness of the proposed method.
- GitHub documentation and demonstration, including the MATLAB code, project description, and experimental results.

Timeline (Weeks 1–13)

Week Task

1–2 Select the project topic and define the objectives and scope of the speech enhancement system.

3–5 Conduct a literature review on STFT, Wiener filtering, and speech enhancement. Collect clean speech samples from LibriSpeech and prepare background noise signals.

6–7 Implement STFT and ISTFT processing in MATLAB and generate noisy speech signals at different SNR levels.

8–9 Implement the Wiener filtering algorithm and integrate it with the STFT-based speech enhancement system. Perform initial testing.

10–11 Evaluate and optimize the system using SNR measurements, waveform comparisons, and spectrogram analysis under different noise conditions.

12–13 Analyse the final results, complete the final report, organise MATLAB code and experimental results.

References

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